Data-driven modelling (4)

Time-series forecasting (short-term traffic predictions)

Nikolai Bode

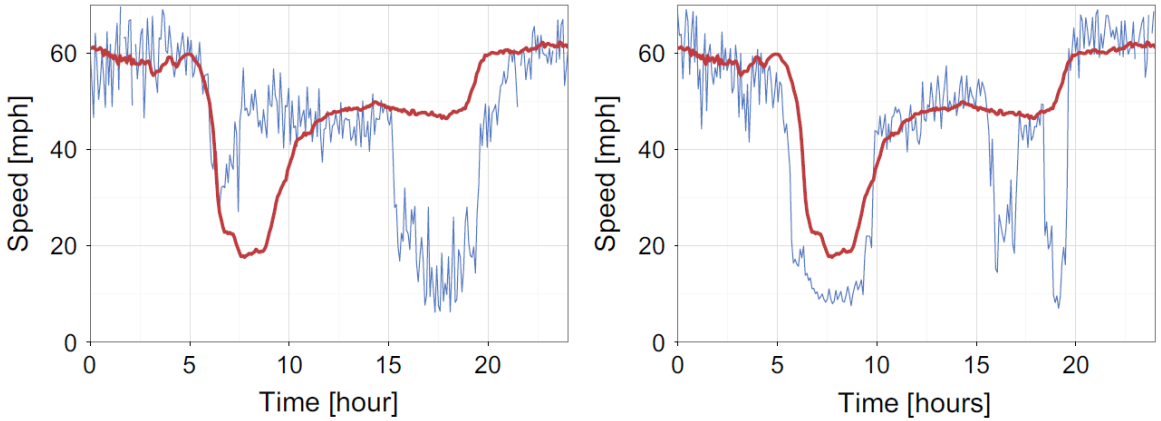
nikolai.bode@bristol.ac.uk

Examples and problem statement

- Time series data is common in transport/traffic, e.g., traffic counts at induction loop, travellers entering railway station/platform, number of people at airport security.
- Forecasting traffic conditions a short time into the future is important for traffic management, and to inform travellers.

Problem statement:

Given $\{X_1, X_2, \ldots, X_t\}$ observed at a location (e.g., speed, flow), predict $X_{t+\Delta}$ for a prediction horizon Δ.

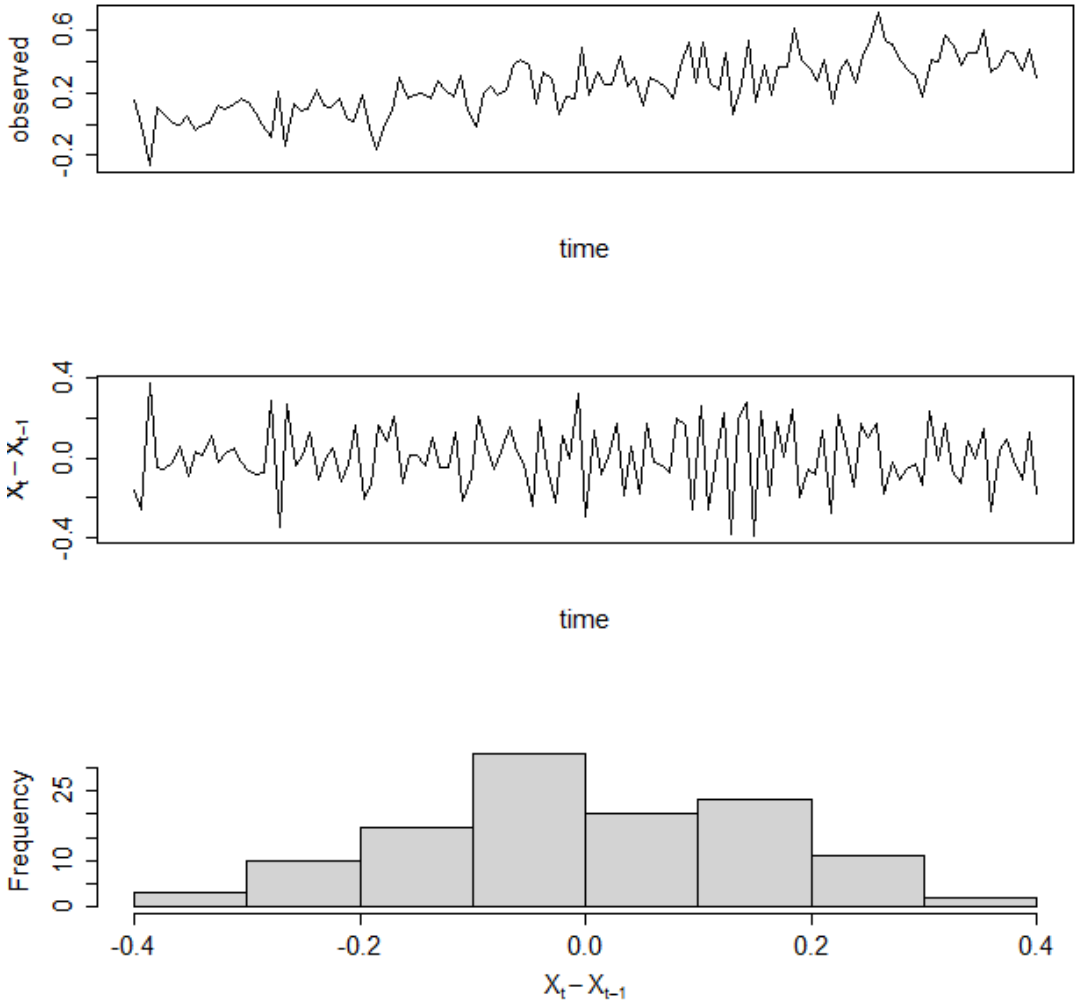


(a) Chicago Bears football game (b) Snow weather

Polson et al. (2017) [3]

Time series properties

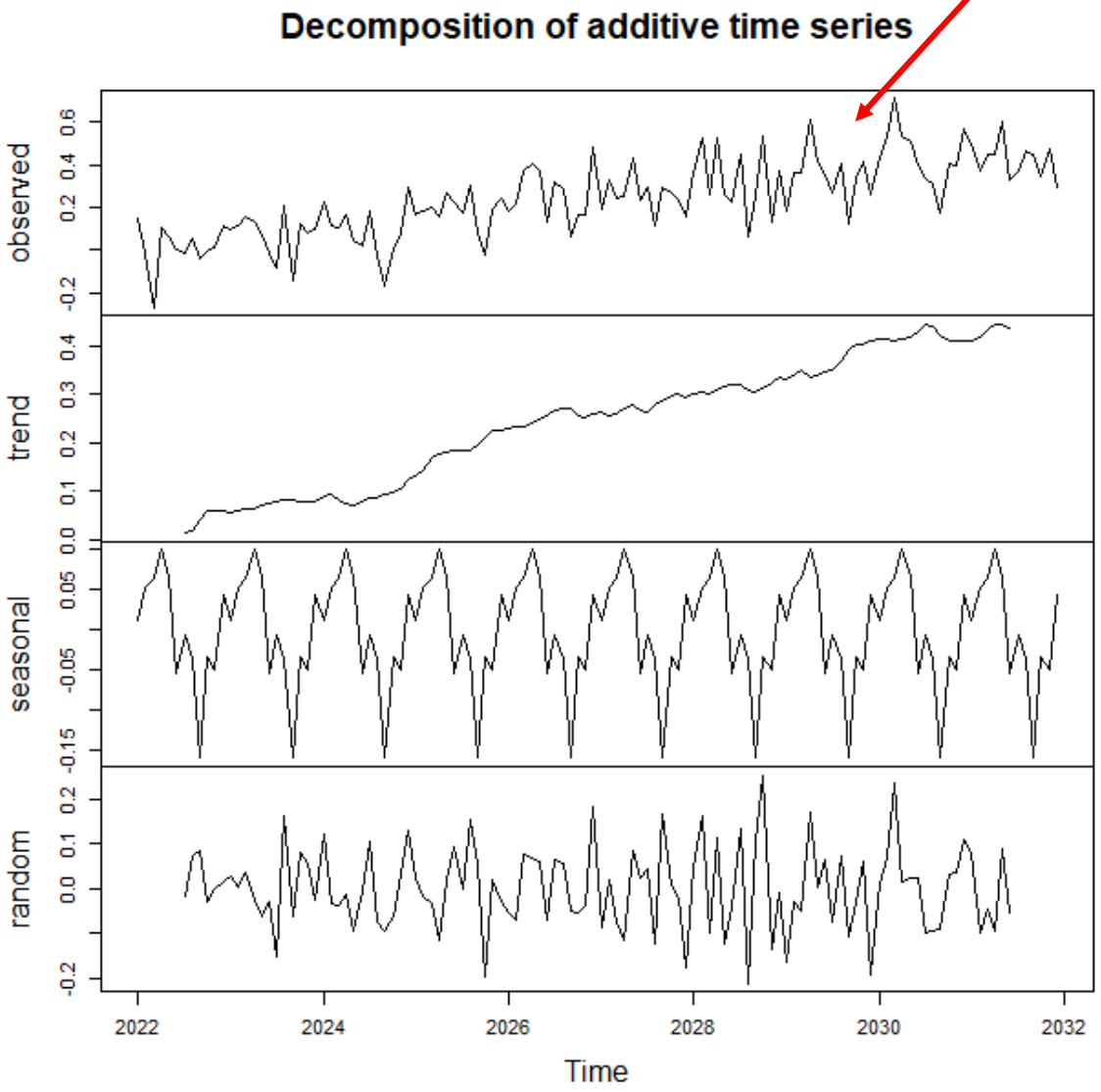
- Need to be considered and can be utilised when modelling time series for forecasting.
- Autocorrelation: correlation(X_i, $X_{i-\delta}$)
  → simplistic random walk model:
  $$(X_i - X_{i-\delta}) \sim N(0, \sigma)$$

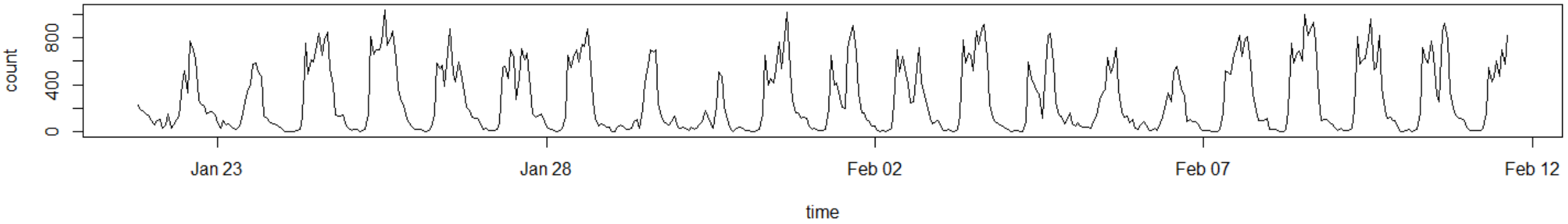


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- Time series decomposition: seasonality, trend, noise
  - Decomposition can additive or multiplicative

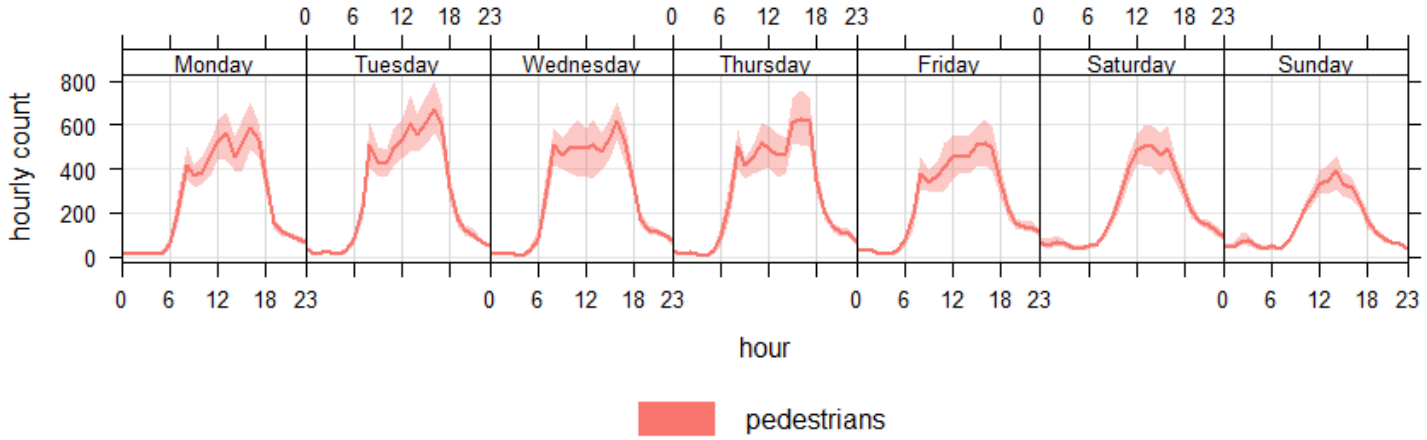
Same time series as on previous slide!

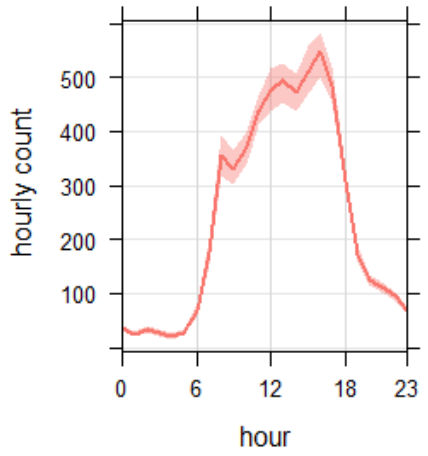


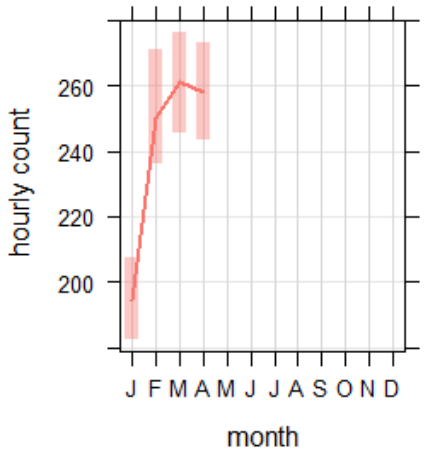


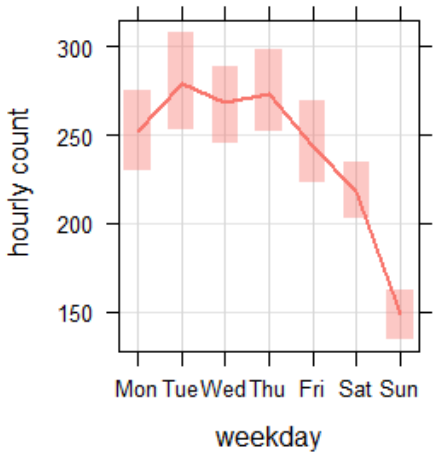
Time series models – structural time-series model

Pedestrian counts at location close to University of Bristol in 2022









mean and 95% confidence interval in mean

Time series models – structural time-series model (STM)

- A model based on the concept of time series decomposition (e.g. [1]).
- $X_t = \mu_t + \gamma_t + \varphi_t + \sum_{i=1}^{m} \sum_{\tau=0}^{q} \omega_{i,t-\tau} y_{i,t-\tau} + \varepsilon_t$

Trend, e.g. $\theta_1 t$

Seasonal component, e.g. $\cos(\theta_2 t)$

Autoregressive component, e.g. $\theta_3 X_{t-1}$

Other explanatory variables, exogeneous variables, e.g. X_t at a different location

Noise or random error component, e.g. $\varepsilon_t \sim N(0, \sigma)$

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- Model fitting is a.g. via Kalman Filters (μ_t and γ_t) and Maximum Likelihood Estimation ($\omega_{i,t-\tau}$ and σ).
- Note: recursive formulations for the components are possible, e.g., trend $\mu_t = \mu_{t-1} + \theta_1$

Time series models – ARMA, ARIMA, SARIMA

- Autoregressive component, $AR(p)$: $X_t = c + \sum_{i=1}^{p} X_{t-i} + \epsilon_t$

  constant → c; Noise → ϵ_t
- AR(0) – noise; AR(1) – random walk

- Moving average component, $MA(q)$: $X_t = \epsilon_t + \sum_{i=1}^{q} \epsilon_{t-i}$

  … allows us to model correlated noise structures.

- ARMA, combine AR and MA: $X_t = c + \sum_{i=1}^{p} X_{t-i} + \sum_{i=1}^{q} \epsilon_{t-i} + \epsilon_t$

Time series models – ARMA, ARIMA, SARIMA

- ARIMA (autoregressive integrated moving average), like ARMA, but with a preprocessing step ("integrated") that removes trend and/or seasonality from the time series using differencing (e.g. replace X_t with $X'_t = X_t - X_{t-1}$). *Need to decide on the number of integration steps.*
- SARIMA (seasonal ARIMA): additional AR and MA terms that are offset by the frequency of the seasonality.

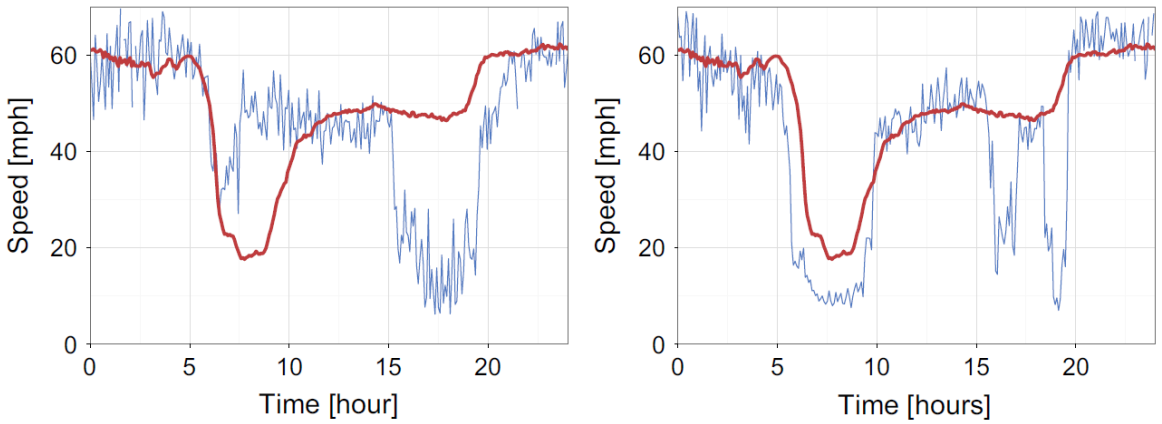
- … and there are more, e.g., SARIMAX, STARIMA; consider exogeneous variables.
- all fitted using Box-Jenkins method.
- ARIMA and friends are less easily interpretable than STM, but they are flexible and capture correlated noise structures [2].

Time series models – machine learning

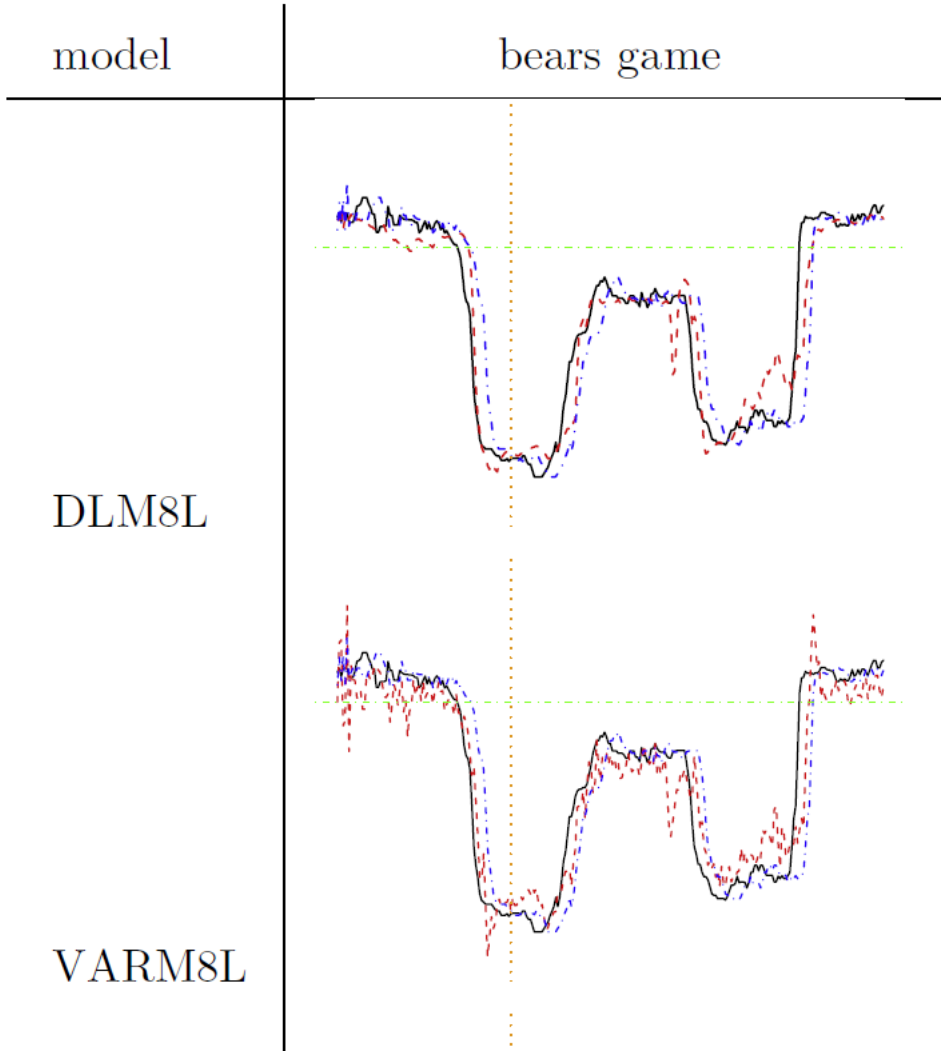
- Short version: replace the predefined structure for a function to predict X_t with a machine learning process (see [2] and [3], for example).
- Examples: Artificial Neural Networks, Support Vector Regression with seasonal kernels, Long Short-Term Memory Networks, Graph Neural Networks (especially for multivariate timeseries)…
- Some claim that machine learning models are the state-of-the-art in traffic forecasting [4], but this debate is currently playing out in research (e.g. https://doi.org/10.48550/arXiv.2301.12603).
- Issues: data accessibility (best data is often owned by corporations), model interpretability, computational costs, robustness (e.g. do changes in road network require re-training?).

Application example 1 [3]

- Forecasting traffic speed in the face of unusual events using machine learning.

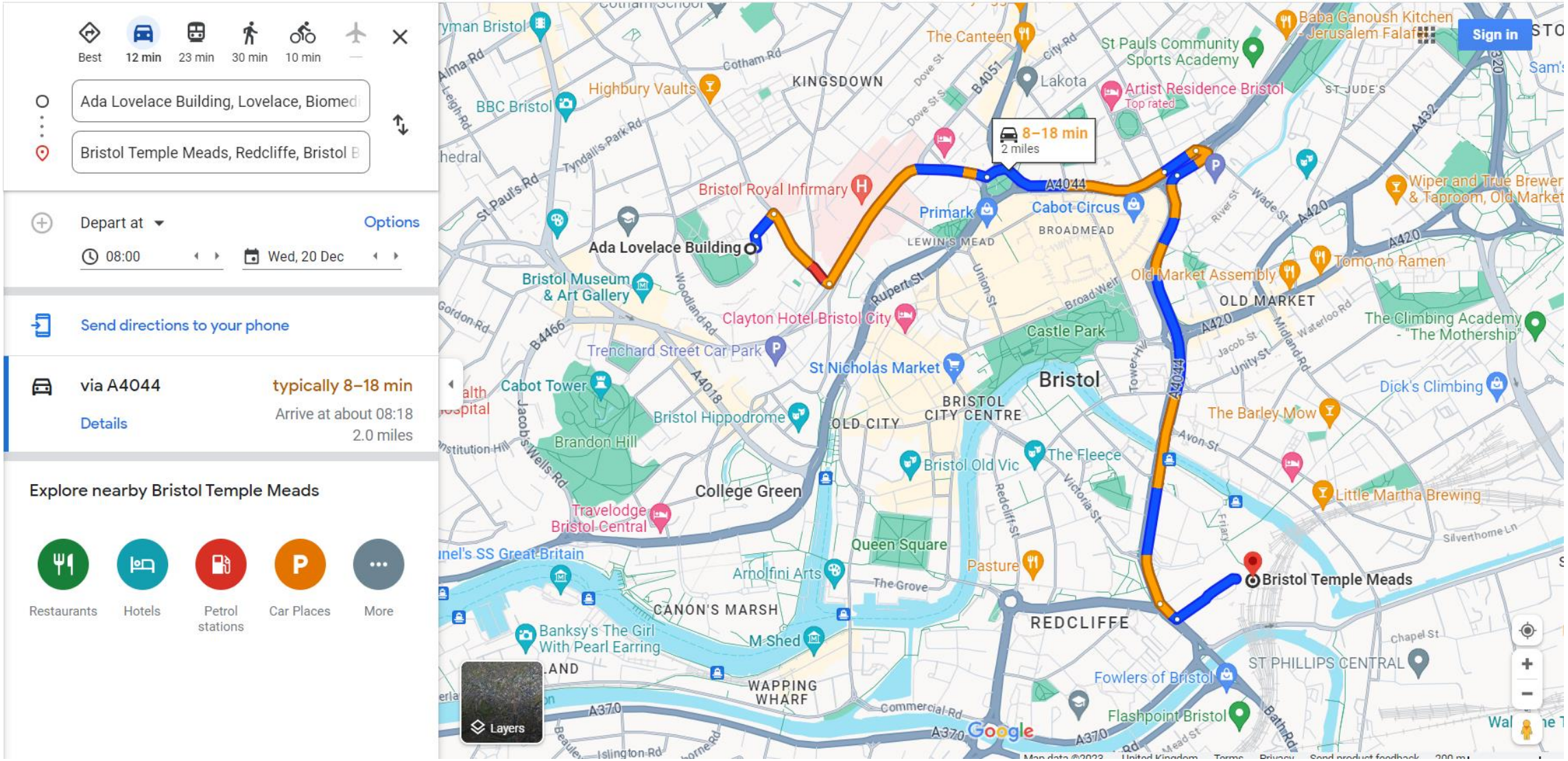


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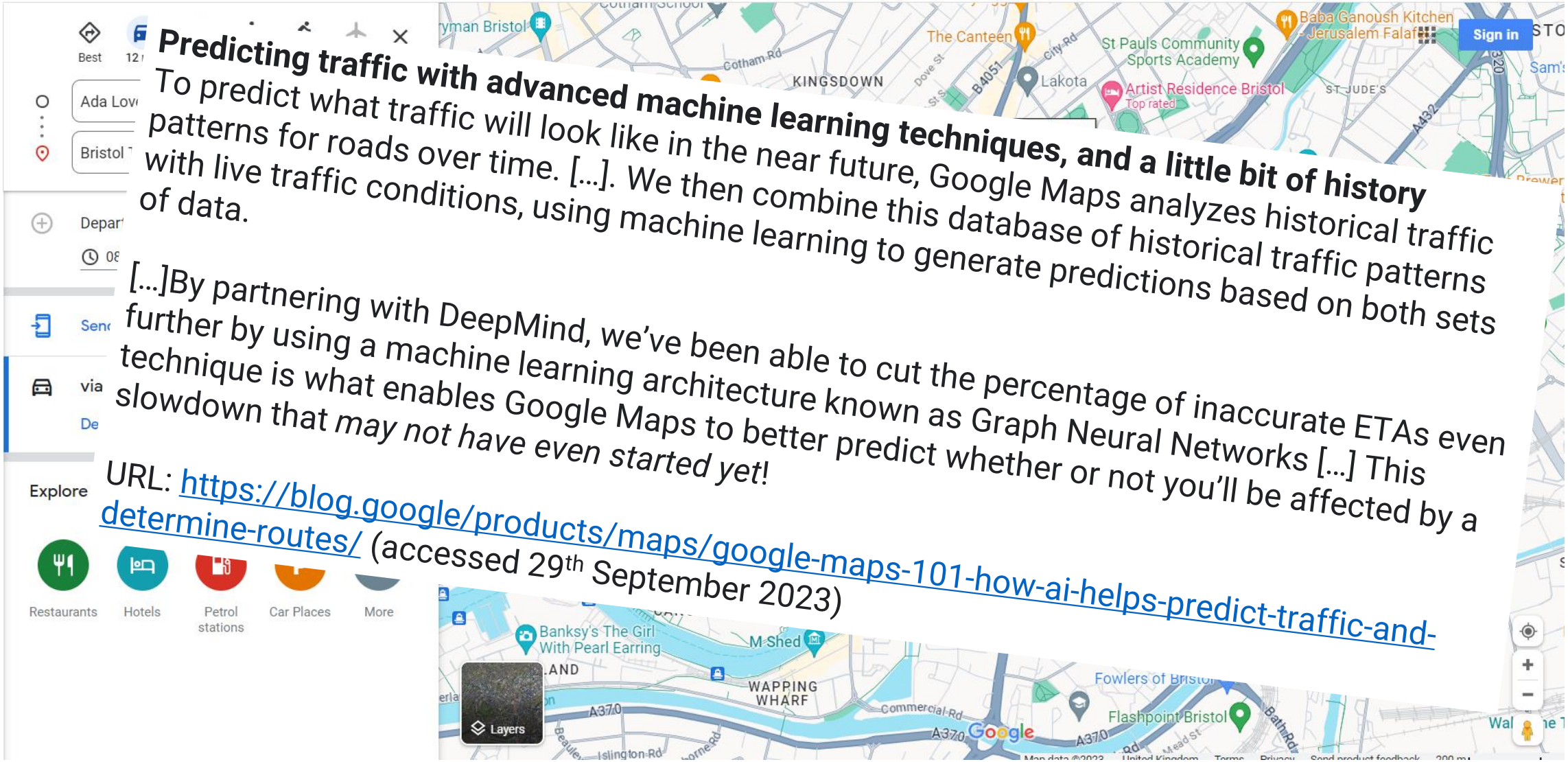


Solid line: original data
Blue line: naïve forecast (basic autoregression)
Red line: machine learning models

Application example 2



Application example 2



Predicting traffic with advanced machine learning techniques, and a little bit of history

To predict what traffic will look like in the near future, Google Maps analyzes historical traffic patterns for roads over time. […]. We then combine this database of historical traffic patterns with live traffic conditions, using machine learning to generate predictions based on both sets of data.

[…]By partnering with DeepMind, we've been able to cut the percentage of inaccurate ETAs even further by using a machine learning architecture known as Graph Neural Networks […] This technique is what enables Google Maps to better predict whether or not you'll be affected by a slowdown that *may not have even started yet!*

URL: https://blog.google/products/maps/google-maps-101-how-ai-helps-predict-traffic-and-determine-routes/ (accessed 29th September 2023)

Summary

- Properties of time series
- Time series models
  - STM
  - ARIMA and friends
  - Machine Learning
- Time series forecasting examples

Further reading

1. Ghosh, B., Basu, B., & O'Mahony, M. (2009). Multivariate short-term traffic flow forecasting using time-series analysis. IEEE transactions on intelligent transportation systems, 10(2), 246-254.
2. Lippi, M., Bertini, M., & Frasconi, P. (2013). Short-term traffic flow forecasting: An experimental comparison of time-series analysis and supervised learning. IEEE Transactions on Intelligent Transportation Systems, 14(2), 871-882.
3. Polson, N. G., & Sokolov, V. O. (2017). Deep learning for short-term traffic flow prediction. Transportation Research Part C: Emerging Technologies, 79, 1-17.
4. Jiang, W., & Luo, J. (2022). Graph neural network for traffic forecasting: A survey. Expert Systems with Applications, 207, 117921.